# AI or Not?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

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WORKBOOK · 1 OF 21

## Recall and choose your starting point

> Record your first explanation before opening notes, then show any correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

**My starting explanation**

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WORKBOOK · 2 OF 21

## Practise with fading help

> Use the photo-library completion task. Keep a trace of the step you completed and any help you used.

Use W2: Complete the photo library explanation for **Evidence** and decision **mechanism**s. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

## Individual reasoning check

> Challenge the proposed claim using the supplied facts before discussing it with a partner.

Use the worked case for **Evidence** and decision **mechanism**s. Proposed explanation: “All rules are outside AI”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved greenhouse case before feedback; Section W4 refers to this same case.

### Independent case: Investigate a new hybrid system

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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## Your learning target

> Use the bus target to explain which component makes each decision and which evidence supports the classification.

A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.

> Would seeing a red display prove that learning occurred?

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WORKBOOK · 6 OF 21

## Solve the given case

> Classify the six supplied devices and support each conclusion with mechanism evidence.

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

> Classify all six using **mechanism evidence**. Which two need **more information**?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

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WORKBOOK · 7 OF 21

## Check a changed case

> Apply the distinction to the toy, photo app and talking clock, and explain each classification.

A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a **learned model**? What **evidence** supports each classification?

My independent answer and reason

**Write or draw your reasoning and evidence.**

**Week 1 Test and reflect**

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WORKBOOK · 8 OF 21

## Design a revealing test

Design a changed or missing-information case and keep the expected result separate from what you observe.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |

| Test record | My evidence |
| --- | --- |
| Why the result occurred | |

WORKBOOK · 9 OF 21

## Project checkpoint

> Justify a project decision and retain the evidence that would let another person check it.

Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or **learned model**; justify the smallest safe solution.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

## Self check

> Assess explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.

- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

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**WORKBOOK · 11 OF 21**

## Transfer practice from the original programme

> Use the original bus question as transfer practice. A previously discussed case is not fresh assessment evidence.

A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.

Would seeing a red display prove that learning occurred?

**My independently reasoned answer**

Record support used: none, hint or teacher explanation.

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**WORKBOOK · 12 OF 21**

## W1 Separate observations from explanations

> Separate what is visible at the door from the question you still need answered about its mechanism.

A door opens when a visitor approaches. Its label says intelligent entrance. List two **observation**s and write one **mechanism** question whose answer would help classify the opening decision.

**My reasoning and evidence**

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## W2 Complete the photo library explanation

> Classify the photo-library decisions and name the records needed to diagnose a misplaced picture.

Use the photo-library case from the lesson. Classify the **prediction**, album-placement rule and owner move. A bicycle photo is in the wrong album. Name the first two records you would inspect.

**My reasoning and evidence**

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## W3 Challenge an unsupported conclusion

> Offer plausible alternatives for the vacuum turn, then identify evidence that could distinguish them.

> Maya says a vacuum must learn because it changes direction after hitting a chair. Give two different **mechanism**s consistent with that **observation**. What **evidence** would distinguish them?

> **My reasoning and evidence**

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## W4 Investigate a new hybrid system

> This is the same greenhouse task as the independent transfer section. Reuse that recorded first attempt.

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

**My reasoning and evidence**

WORKBOOK · 16 OF 21

## W5 Design an evidence card

> Design one product with two mechanisms, then state one supported and one unsupported claim.

Invent one product containing two **mechanism**s. State the supplied facts, draw an information path and write one claim the facts support and one they do not. Do not use real classmates' personal information.

**My reasoning and evidence**

WORKBOOK · 17 OF 21

## Feedback, return check and learning record

> Record the feedback, the first step you repaired and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 18 OF 21**

## Pictorial case practice: AI or Not?

> Explain the umbrella picture and predict a changed boundary using the supplied facts.

A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

A weather model was fitted on earlier weather records; its forecast is **70%** rain.

A **written rule** shows TAKE ONE at rain forecasts of **60%** or more. The teacher decides whether to distribute umbrellas.

AI ACADEMY / AI-1/1

# AI or Not?

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Weather records

Earlier observations were used to fit the forecasting model.



### 2. Forecast

The learned model outputs 70% rain, not a guaranteed shower.



### 3. Display rule

70 is at least 60, so the written rule displays TAKE ONE.



### 4. Final decision

The teacher checks current conditions before distributing umbrellas.

**CHANGE ONE THING**

Keep the forecast but raise the display boundary to 80%.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Point to the learned stage. Does the changed display prove the weather forecast changed?

**My picture-based explanation and changed-case prediction**

### Record the help you used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

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## Project portfolio milestone

Keep the sorter milestone, input, deciding step, result and limitation together in the project record.

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## Supported practice and learning evidence

> Use the supported questions once, compare with feedback after attempting them, and record what you repaired.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? The teacher checks current conditions before distributing umbrellas.

1. Weather records

2. Final decision

3. Display rule

4. Forecast

**My selected conclusion, deciding fact and explanation**

### Hint · use only after trying

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after your attempt

After your attempt, compare with the supported explanation: Final decision The teacher checks current conditions before distributing umbrellas. The **fitting** record supports calling the forecasting stage learned.

Comparing 70 with 60 is an explicit rule. The final action belongs to a person. Merely seeing TAKE ONE would not reveal any of these **mechanism**s; the **process** notes supply that **evidence**.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Keep the forecast but raise the display boundary to **80%**. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The display no longer says TAKE ONE.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · use only after trying

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after your attempt

After your attempt, compare with the supported explanation: The display no longer says TAKE ONE. 70 is below 80; changing a display rule does not refit the model.

#### Repair a missing building block

This is the starting lesson. Ask your teacher to model one **observation**, one **written rule** and one deciding step, then try your own example.

#### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

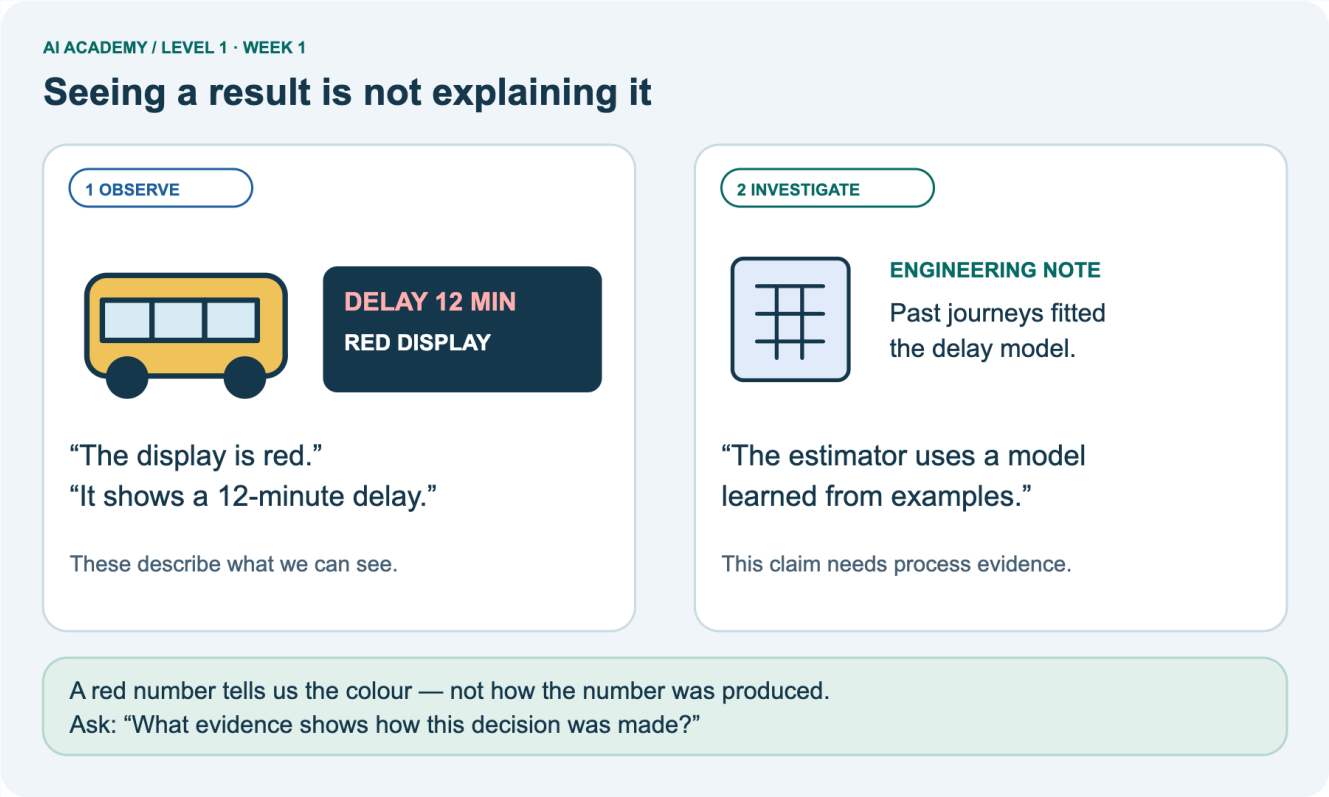
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WORKBOOK · 21 OF 21

## Visual concept journey

> Explain each picture using its labels and evidence, then record a relevant limit.

## 01. Seeing a result is not explaining it



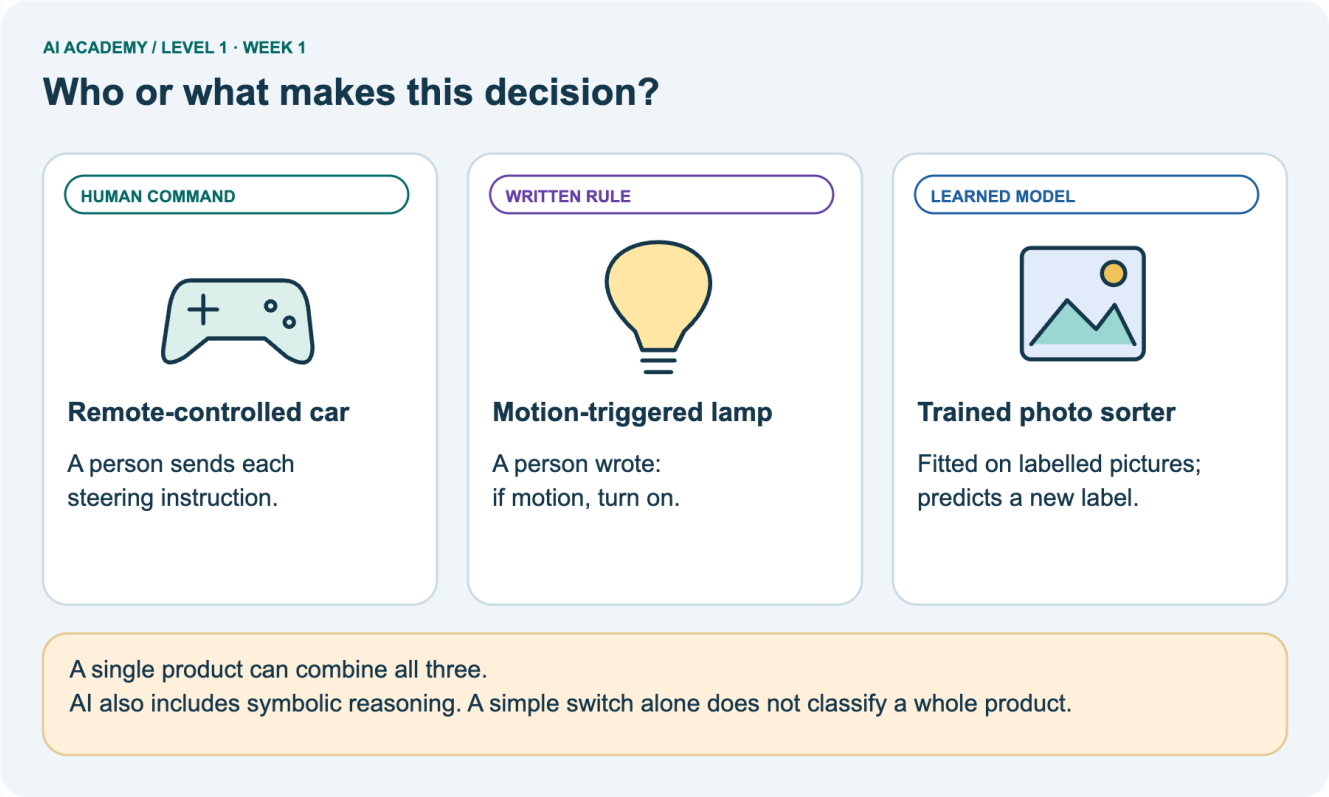

Follow the labelled picture with the matching explanation. Open the vector diagram

A bus has a red display showing a 12-minute delay. That is an **observation**. A supplied engineering note says past journeys fitted the delay model; that is **evidence** about its **mechanism**.

> Name one **observation** and one claim that needs more **evidence**.

### My picture explanation, deciding evidence and one limit

## 02. Who or what makes this decision?



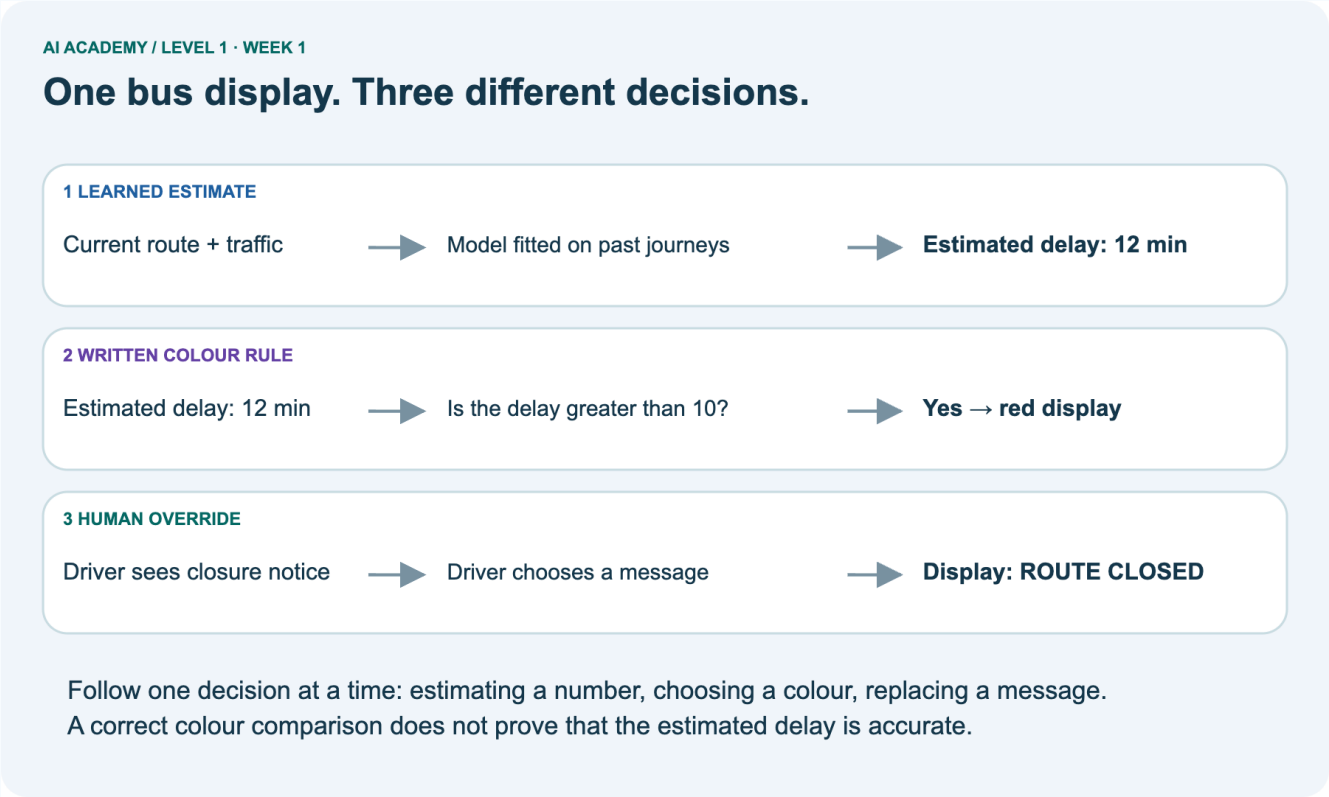

Follow the labelled picture with the matching explanation. Open the vector diagram

A remote-controlled car follows human commands, a motion lamp follows a **written rule**, and a trained photo sorter applies a **learned model**. A product can combine **mechanism**s. AI also includes symbolic reasoning.

> Point to who or what makes each decision.

### My picture explanation, deciding evidence and one limit

## 03. One bus display. Three different decisions.



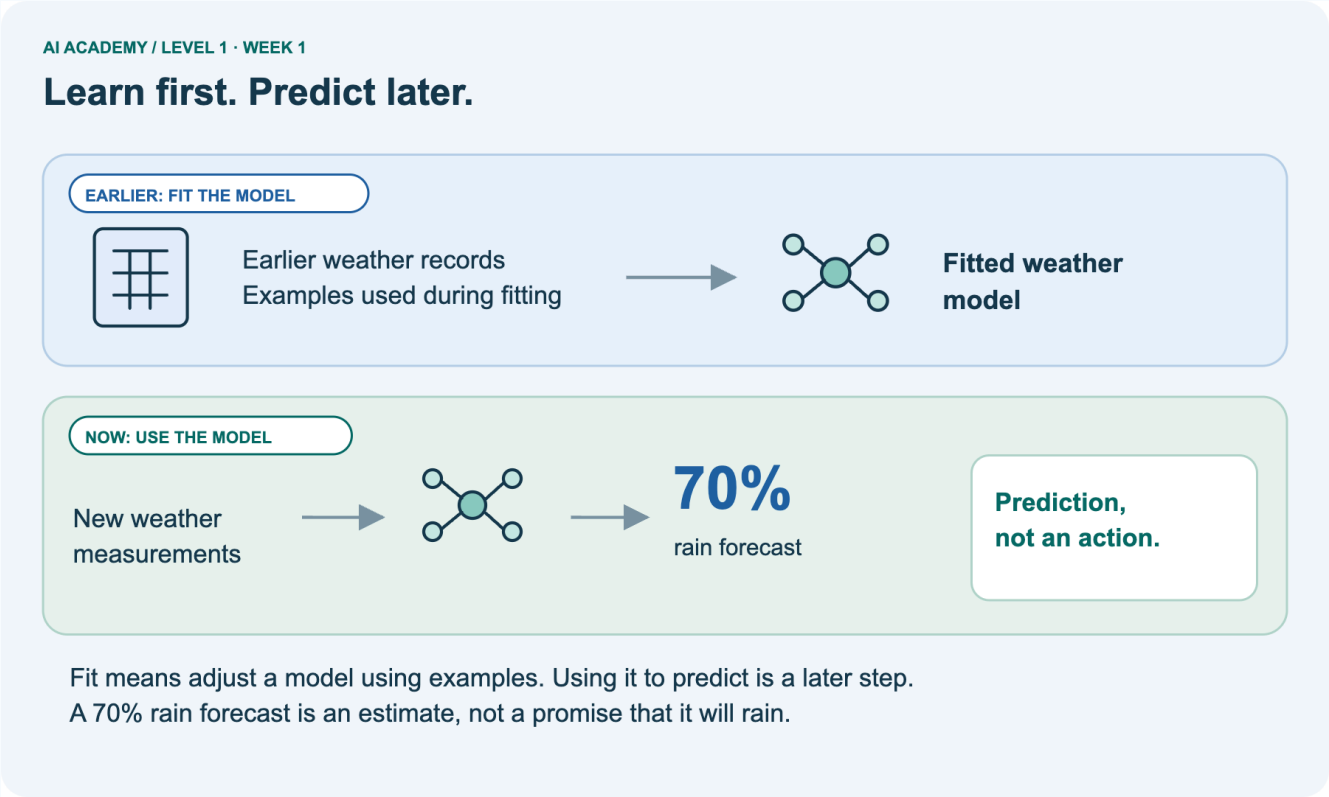

Follow the labelled picture with the matching explanation. Open the vector diagram

Three separate rows trace the bus estimate, display colour and driver override. The fitted model estimates **12 minutes**; the written greater-than-10 rule makes the display red; the driver can choose ROUTE CLOSED.

> Trace the number, colour and override separately.

### My picture explanation, deciding evidence and one limit

## 04. Learn first. Predict later.



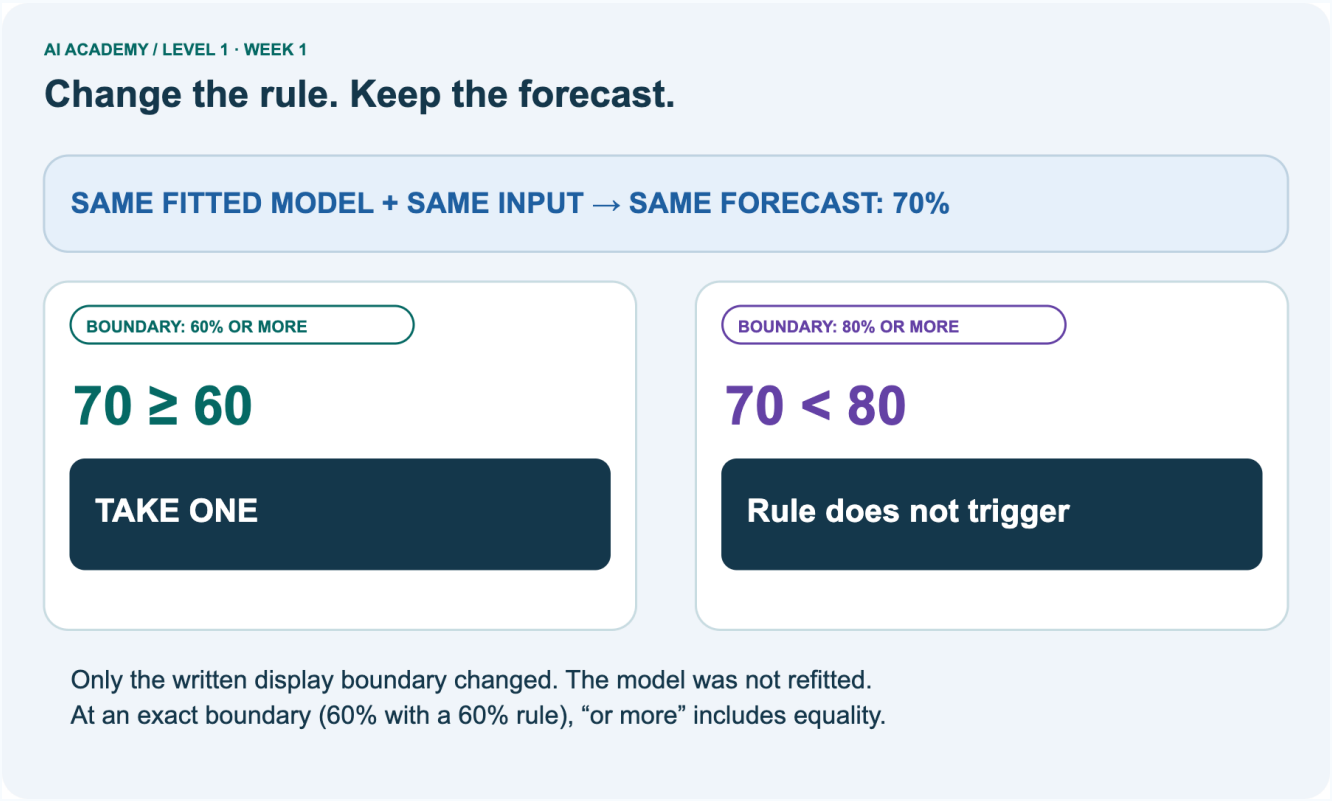

Follow the labelled picture with the matching explanation. Open the vector diagram

Earlier weather records are used to fit a model. Later, new measurements are passed to that fitted model to obtain a 70 percent rain forecast. It is an estimate, not a guarantee.

> Which steps use earlier examples, and which use a fitted model?

### My picture explanation, deciding evidence and one limit

### 05. Change the rule. Keep the forecast.




Follow the labelled picture with the matching explanation. Open the vector diagram

With a 70 percent rain forecast, a 60 percent-or-more rule displays TAKE ONE. An 80 percent-or-more rule does not trigger that message. Only the boundary changed; the fitted model and forecast did not.

> Keep the forecast fixed at **70%**. What changes when the boundary becomes **80%**?

**My picture explanation, deciding evidence and one limit**

## 06. “Not enough information” is a useful answer

AI ACADEMY / LEVEL 1 · WEEK 1

### “Not enough information” is a useful answer



**“Smart” robot vacuum**

It moves and cleans.
How does it choose a route?
The description does not say.

ASK FOR MECHANISM EVIDENCE

Person steering? → evidence of human control
Explicit route instructions? → evidence of rules
Model fitted on examples? → evidence of learning

**Several may be combined.**
**If evidence is missing, say:**
**“I need more information.”**

PROJECT: Draw input → deciding step → output for your classroom sorter.
Label each deciding step: human / written rule / learned model / not yet known.
Then name the evidence for your label.

Follow the labelled picture with the matching explanation. Open the vector diagram

An unspecified smart robot vacuum moves and cleans, but its route **mechanism** is unknown. Ask for **evidence** of human commands, **written rule**s or model **fitting**, and allow combinations. Apply this reasoning to a classroom sorter.

Which **evidence** would change your answer about the vacuum?

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.